

Session Type State Spaces Form Lattices

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We prove that the state space of every well-formed session type, quotiented by strongly connected components, forms a bounded lattice; n -ary parallel composition yields product lattices. Two consequences follow: duality preserves the lattice up to isomorphism, and Gay–Hole width subtyping corresponds to lattice embedding for non-recursive types. We validate this on 108 benchmark protocols across networking, databases, distributed systems, AI, and fault tolerance: all form lattices, 93 distributive, 15 non-distributive. Mechanised in Lean 4 with two independently developed tool implementations.

1 Introduction

Session types [15, 32] describe the communication protocols that govern interaction with objects, channels, or services. A session type specifies the allowed sequences of operations — method calls, branches, selections — together with their ordering constraints. The *state space* of a session type is a labelled transition system (LTS) *over states* obtained by “executing” the type: each state represents a protocol stage, and each transition a permitted action. The semantics is specification-level rather than operational: states are the unit of reachability, not reduction on terms.

State spaces are routinely used for verification, subtype checking, and protocol analysis, yet their algebraic structure has received little attention. Gay and Hole [13], for instance, study subtyping as a preorder on session types but do not investigate the structure of the underlying state spaces. In this paper we show that *the state space of every well-formed session type, after quotienting by strongly connected components, is a bounded lattice*. We call this lattice a *reticulate* — from Latin *reticulatus*, the root of “lattice”, giving session type lattices their own name while remaining etymologically precise.

The lattice structure arises naturally from the session type constructors: branch ($\&$) and selection ($\oplus$) create shared extrema that serve as meets and joins; recursion introduces cycles absorbed by the *strongly-connected-component (SCC) quotient* — the standard graph-theoretic operation that collapses each set of mutually reachable states into one equivalence class, yielding a directed acyclic graph (DAG); and n -ary parallel composition ($\parallel$) yields n -dimensional product lattices. The $\parallel$ constructor is where lattice structure becomes *necessary* rather than merely convenient. Without $\parallel$, the SCC quotient of a session type’s state space is a tree (pure sequencing), a diamond (binary choice), or a DAG (nested choice). With $\parallel$, the n -fold product of state spaces inherently has meets and joins corresponding to fork and synchronisation points of concurrent execution.

Our contributions are: (1) the *Reticulate Theorem*: the SCC quotient of every well-formed session type’s state space is a bounded lattice, by structural induction (Section 3); (2) *duality preservation*: swapping branch and selection preserves state-space isomorphism (Section 4.1); (3) *subtyping as lattice embedding*: Gay–Hole width subtyping corresponds to lattice embedding for non-recursive types, with the selection case derived from the branch case via duality (Section 4.2); (4) *validation* (Section 6): an empirical study on 108 benchmark protocols using two independently developed tools; mechanised verification of the proof in Lean 4 (15 modules, zero sorry); and a distributivity classification: N_5 from

depth-unequal choice, M_3 from wide choice ($n \geq 3$), which together account for all 15 non-distributive cases.

Section 2 recalls the relevant background. Section 3 defines the state-space construction and presents the proof. Section 4 develops the duality and subtyping applications. Section 5 demonstrates each result on a concrete concurrent file protocol. Section 6 reports the implementation, the Lean 4 mechanisation, and the empirical study on 108 protocols. Section 7 discusses related work and Section 8 concludes.

2 Preliminaries

2.1 Session Types

The grammar below targets method-call protocols on a single shared *object*; channel-based bi- and multiparty communication (send/receive, delegation, as in process calculi [15]) is out of scope unless the channel is itself modelled as an object, in which case the present framework applies directly. We restrict attention to finite-state (regular) session types.

The grammar below extends typestate session types [14, 33, 19] with parallel composition, letting the lattice construction reach objects shared by concurrent clients:

$$\begin{array}{ll}
 S ::= & \&\{m_1 : S_1, \dots, m_n : S_n\} & \text{(branch, } n \geq 0) \\
 & | \oplus\{l_1 : S_1, \dots, l_n : S_n\} & \text{(selection, } n \geq 0) \\
 & | S_1 \parallel \dots \parallel S_n & \text{(parallel, } n \geq 2) \\
 & | \mu X. S & \text{(recursion)} \\
 & | X & \text{(variable)} \\
 & | \mathbf{end} & \text{(termination)} \\
 D ::= & S \mid S, X_1 = S_1, \dots, X_n = S_n & \text{(declaration, equations } n \geq 0)
 \end{array}$$

A declaration D is a root session type S optionally followed by named equations $X_i = S_i$. The equations decompose a protocol into named, reusable phases; cyclic references express recursion (including mutual recursion), subsuming $\mu X. S$ as the special case $X = S$. We retain $\mu X. S$ as a separate primitive, equivalent to the single-equation declaration $X = S$.

The branch $\&\{m_1 : S_1, \dots, m_n : S_n\}$ represents a state in which the object unconditionally offers n method calls; the client exercises an external choice of which to invoke, and the protocol continues as S_i for the chosen m_i ($n = 0$ is the empty offer). In contrast, the selection $\oplus\{l_1 : S_1, \dots, l_n : S_n\}$ represents an internal choice in which the object returns one of n enumeration labels l_i ; the client must dispatch on the returned label and proceed with the corresponding continuation S_i ($n = 0$ is the empty internal choice). Parallel composition $S_1 \parallel \dots \parallel S_n$ is the flat n -ary form (no nesting), with each S_i describing one client's interactions with the shared object; the sub-protocols run concurrently and may cooperate through that object, with clients playing dual or complementary roles (e.g., reader/writer, publish/subscribe). A variable X is a reference to its binder, either a $\mu X. \cdot$ form or an equation $X = S$.

The constant **end** is the explicit termination primitive, not synonymous with the empty branch $\&\{\}$: $\&\{\}$ merely denotes a state with no enabled methods. Every terminated session has no methods, but the converse fails — a parallel branch that has exhausted its actions has no methods to offer yet is not “terminated” until its siblings complete. We retain **end** as a primitive to preserve this asymmetry, matching the additive units of classical linear logic [34]: $\&\{\} \cong \top$, $\oplus\{\} \cong \mathbf{0}$, dual via $\mathbf{0}^\perp = \top$.

Definition 1 (Well-formedness). A session type (or declaration) is *well-formed* if all of the following hold.

1. **Closedness.** S (resp. D) has no free variables, where binders are $\mu X.\cdot$ and equation left-hand sides X .
2. **Contractiveness** (also called *guardedness* in the process-calculus tradition). Every recursion $\mu X.S$ has at least one constructor between $\mu X.\cdot$ and each occurrence of X in S ; in equation systems, every cyclic reference path passes through at least one constructor.
3. **Termination.** From every state, there is a path to **end**.
4. **Parallel closedness.** In every parallel composition $S_1 \parallel \dots \parallel S_n$ occurring in S , no variable bound outside the parallel occurs free in any S_i .

Closedness, contractiveness, and parallel closedness are standard in the session-type literature [15, 32, 19, 14] and rule out genuinely ill-formed types — dangling references, uncontractive recursions like $\mu X.X$, and parallel branches that capture outer recursion variables (Lemma 14). Termination is specific to our setting and characterises which types form bounded lattices (Theorem 15), not which are admissible: non-terminating types remain typeable, with both tools exposing identical `--non-termination` handling (Section 6).

2.2 Lattice Theory

We recall the standard lattice-theoretic vocabulary used in the proofs of Theorem 15; see [4] for a textbook treatment.

Bounded lattice. A bounded lattice is a poset $(L, \leq)$ with a least element $\perp$ and a greatest element $\top$ such that every pair of elements has a *meet* (greatest lower bound) $a \wedge b$ and a *join* (least upper bound) $a \vee b$.

Product lattice. Given bounded lattices $(L_1, \leq_1), \dots, (L_n, \leq_n)$, their product $L_1 \times \dots \times L_n$ is ordered componentwise: $(a_1, \dots, a_n) \leq (b_1, \dots, b_n)$ iff $a_i \leq_i b_i$ for all i . Meets, joins, top, and bottom are componentwise, and the product of finitely many bounded lattices is itself a bounded lattice.

Distributive lattice. A bounded lattice is *distributive* if $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ holds for all elements; equivalently, if it contains no sublattice isomorphic to N_5 (the pentagon) or M_3 (the diamond with three atoms).

Upward-closed subset. A subset D of a poset $(L, \leq)$ is *upward-closed* if $x \in D$ and $x \leq y$ imply $y \in D$; it is *proper* when $D \neq L$. The dual notion is *downward-closed*.

SCC quotient. A *strongly connected component* (SCC) of a directed graph is a maximal set of mutually reachable vertices; the *SCC quotient* collapses each SCC to a single node, yielding a DAG [29]. In our setting, SCCs collect the states a recursive protocol can revisit; on the quotient, reachability defines a partial order, lifted to a bounded lattice in Lemma 12.

3 State Space Construction and the Reticulate Theorem

Definition 2 (State space). A state space is a tuple $(Q, \Sigma, \delta, q_\top, q_\perp)$ where Q is a finite set of states, Σ is a finite alphabet of action labels, $\delta \subseteq Q \times \Sigma \times Q$ is the transition relation, $q_\top \in Q$ is the initial state, and $q_\perp \in Q$ is the terminal state. The construction of Definition 3 assigns a state space $\mathcal{L}(S)$ to each well-formed session type S .

Definition 3 (State-space construction $\mathcal{L}(S)$). The state space $\mathcal{L}(S) = (Q, \Sigma, \delta, q_\top, q_\perp)$ is defined by structural induction on S . The induction yields state spaces, not bounded lattices: $\mathcal{L}(S)$ carries *placeholder* states for free variables, eliminated by cases (4) and (5) below via back-edge redirection. For closed well-formed S , the SCC quotient $\mathcal{L}(S)/\equiv$ is a bounded lattice (Definition 6 and Theorem 15).

1. **End.** $\mathcal{L}(\text{end}) = (\{q_0\}, \emptyset, \emptyset, q_0, q_0)$ — a single state, both initial and terminal.
2. **Branch** $\&\{m_i : S_i\}_{i=1..n}$.

$$\begin{aligned} Q &= \{q_0\} \cup \bigcup_i (Q_i \setminus \{q_\perp^i\}) \cup \{q_\perp\}, \\ \Sigma &= \{m_1, \dots, m_n\} \cup \bigcup_i \Sigma_i, \\ \delta &= \{(q_0, m_i, q_\top^i)\}_i \cup \bigcup_i \delta_i[q_\perp^i := q_\perp], \end{aligned}$$

where $Q_i, \Sigma_i, \delta_i, q_\top^i, q_\perp^i$ are the components of $\mathcal{L}(S_i)$; $\delta_i[q_\perp^i := q_\perp]$ denotes δ_i with every $q_\perp^i$ replaced by the shared $q_\perp$; $q_\top = q_0$ and $q_\perp$ is shared across children. For $n = 0$, $Q = \{q_0, q_\perp\}$, $\Sigma = \{\tau\}$, $\delta = \{(q_0, \tau, q_\perp)\}$ with τ a silent action for vacuous termination; the SCC quotient is the chain $q_0 > q_\perp$.

3. **Selection** $\oplus\{l_i : S_i\}_{i=1..n}$. Identical tuple to branch at the state-space level: the construction depends only on the continuations (the sub-types S_i that describe what the protocol does *after* taking the i -th label), not on polarity. External vs. internal choice ($\&$ vs. $\oplus$) matters for subtyping [13] and duality but not for reachability.
4. **Recursion** $\mu X. S'$. Build $\mathcal{L}(S') = (Q', \Sigma', \delta', q_\top', q_\perp')$ with fresh placeholder states for each occurrence of X ; let $P_X \subseteq Q'$ collect these placeholders. Then $Q = Q' \setminus P_X$, $\Sigma = \Sigma'$, $\delta = \delta'[P_X := q_\top']$ (redirect every transition targeting a placeholder to $q_\top'$, creating back-edges), $q_\top = q_\top'$, $q_\perp = q_\perp'$.
5. **Named definitions** $D = S, X_1 = S_1, \dots, X_k = S_k$ (root session type followed by $k \geq 0$ equations). Build $\mathcal{L}(S)$ and each $\mathcal{L}(S_j)$ with placeholders $p_{i,j}$ for each occurrence of X_j . For each name X_j , replace every transition targeting any $p_{*,j}$ with a transition to $q_\top^{S_j}$, and remove the placeholders. The result is $\mathcal{L}(D) = (Q, \Sigma, \delta, q_\top^S, q_\perp^S)$ where $Q = (Q^S \cup \bigcup_j Q^{S_j})$ with placeholders removed, $\Sigma = \Sigma^S \cup \bigcup_j \Sigma^{S_j}$, δ is the global redirection. Acyclic name references are inlined; cyclic references create back-edges absorbed by the SCC quotient (formally Definition 6 below) exactly as in single-variable recursion.
6. **Parallel** $S_1 \parallel \dots \parallel S_n$. $\mathcal{L}(S_1 \parallel \dots \parallel S_n) = (Q_1 \times \dots \times Q_n, \Sigma, \delta_\times, (q_\top^1, \dots, q_\top^n), (q_\perp^1, \dots, q_\perp^n))$ with $\Sigma = \bigcup_i \Sigma_i$ and $\delta_\times$ the componentwise advance relation: $(s_1, \dots, s_i, \dots, s_n) \xrightarrow{a} (s_1, \dots, s'_i, \dots, s_n)$ for each $s_i \xrightarrow{a} s'_i$ in δ_i .

Definition 4 (Reachability). Write $s_1 \rightarrow s_2$ (“ s_1 reaches s_2 ”) iff there exists a path from s_1 to s_2 in (Q, δ) .

Proposition 5 (Preorder). The relation $\rightarrow$ is reflexive and transitive, hence a preorder; cycles from recursion prevent antisymmetry.

Definition 6 (SCC quotient and its order). The *SCC quotient* $\mathcal{L}(S)/\equiv$ identifies mutually reachable states (those in the same protocol phase), yielding the canonical antisymmetric quotient of $\mathcal{L}(S)$. We order it by $[s_1] \leq [s_2]$ iff $s_2 \twoheadrightarrow s_1$: a state is *smaller* when less protocol remains, so execution descends from $\top$ to $\perp$. The existence of unique extrema $q_\top$ and $q_\perp$ is Proposition 7 below.

Proposition 7 (Unique extrema). *For every well-formed S , $\mathcal{L}(S)$ has a unique initial state $q_\top$ that reaches all states, and a unique terminal state $q_\perp$ reachable from all states.*

Proof. By structural induction on S : each constructor explicitly designates the extrema, and the back-edges introduced by recursion preserve them in the SCC quotient. $\square$

Each constructor of Definition 3 corresponds to a lattice operation on its sub-state-spaces: **end** yields the one-element lattice; choice glues sub-lattices through a fresh top into a diamond; recursion absorbs an upward-closed set into its body's top via the SCC quotient; named definitions iterate that absorption; parallel takes a Cartesian product. The proof of the Reticulate Theorem (Theorem 15, Section 3.5) walks this grammar of lattice-builders by structural induction on well-formed S , with one lemma per constructor. The recursion case (Lemma 12) inducts over bodies-with-placeholders, with the IH carrier specified in the lemma's hypotheses. Proof sketches appear inline; full proofs of every result are in Appendix A.

3.1 End

Lemma 8 (End). $\mathcal{L}(\mathbf{end})/\equiv$ is a bounded lattice.

Proof. $\mathcal{L}(\mathbf{end})$ is a single state q with $q_\top = q_\perp = q$; the SCC quotient is the one-element poset $\{[q]\}$, with $[q]$ as both top and bottom. All lattice axioms hold trivially. $\square$

3.2 Choice and Sequencing

Branch and selection share their state-space structure: we establish this first, then treat both polarities uniformly as $C\{a_1 : S_1, \dots, a_n : S_n\}$.

Lemma 9 (Polarity erasure). *For any labels $a_1, \dots, a_n$ and well-formed types $S_1, \dots, S_n$,*

$$\mathcal{L}(\&\{a_1:S_1, \dots, a_n:S_n\}) = \mathcal{L}(\oplus\{a_1:S_1, \dots, a_n:S_n\}).$$

Proof. Both constructions create a fresh $q_\top$ with n transitions to sub-state-space entries, sharing $q_\perp$ — depending only on the continuations, not on the polarity. Polarity is a typing discipline, not a structural property of the state space. $\square$

Lemma 10 (Choice). *If $\mathcal{L}(S_i)/\equiv$ is a bounded lattice for each i , then $\mathcal{L}(C\{a_1 : S_1, \dots, a_n : S_n\})/\equiv$ is a bounded lattice.*

Proof. The construction creates a fresh $q_\top$ with transitions to each sub-lattice entry above a shared $q_\perp$. For states in the same sub-lattice, meets and joins exist by induction. For states in different sub-lattices, $q_\top$ is the only common upper bound and $q_\perp$ the only common lower bound (Figure 1). $\square$

The syntactic sugar $m.S \equiv \&\{m : S\}$ is the $n = 1$ case of Lemma 10: the fresh $q_\top$ has a single transition to $q_\top^S$, yielding a chain that prepends one element above the existing lattice. No separate lemma is needed.

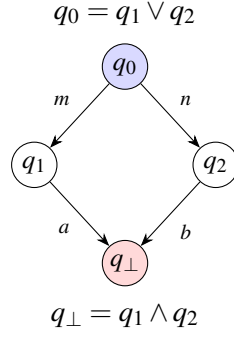


Figure 1: State space of $\{m : a . \mathbf{end}, n : b . \mathbf{end}\}$: the diamond lattice M_2 . The initial state is the join, the terminal state the meet, of q_1 and q_2 .

3.3 Recursion

Recursion is the most delicate case: back-edges create cycles that cause the SCC quotient to merge classes.

Lemma 11 (Top absorption). *Let $(L, \leq)$ be a bounded lattice and D a proper upward-closed subset of L containing $\top$ (equivalently, $\perp \notin D$). Define $L' = (L \setminus D) \cup \{\top\}$ with the restricted ordering. Then L' is a bounded lattice.*

Proof. Since D is upward-closed and proper, $\perp \notin D$, so L' retains both extrema. Meets in $L \setminus D$ stay in $L \setminus D$ (the complement of an upward-closed set is downward-closed) and are preserved. Joins in $L \setminus D$ are likewise preserved, except those whose L -join lies in D , which collapse to $\top$ in L' — any non- $\top$ upper bound would itself lie in D by upward-closure. $\square$

Lemma 12 (Recursion). *Let S be a body with a single free variable X , such that the state space $\mathcal{L}(S)$ with placeholder state p_X satisfies: (i) $q_\top^S$ reaches every non-placeholder state; (ii) the placeholder-free SCC quotient is a bounded lattice; and (iii) the recursion $\mu X.S$ satisfies the **Termination** clause of Definition 1. Then $\mathcal{L}(\mu X.S)/\equiv$ is a bounded lattice.*

Proof. Adding back-edges from each placeholder p_X to $q_\top^S$ collapses these states with $q_\top^S$ into one SCC class. The set D of quotient classes absorbed into $[q_\top^S]$ is upward-closed (mutual reachability through $q_\top^S$ propagates upward), and the Termination clause (Definition 1) ensures $[q_\perp] \notin D$, making D a proper upward-closed subset of the body quotient. Lemma 11 then yields the lattice. $\square$

Lemma 13 (Named definitions). *Let $D = S, X_1 = S_1, \dots, X_k = S_k$ be a well-formed declaration. If $\mathcal{L}(S)/\equiv$ and each $\mathcal{L}(S_i)/\equiv$ are bounded lattices, then $\mathcal{L}(D)/\equiv$ is a bounded lattice.*

Proof. Acyclic name references inline trivially. Cyclic references create back-edges that merge states into SCCs, exactly as in Lemma 12; iterated application of Lemma 11 (one step per SCC) yields a bounded lattice. The SCC quotient depends only on the reachability graph, not on the order in which absorptions are applied, so the construction is well-defined up to lattice isomorphism (a confluence property of absorption-step composition). $\square$

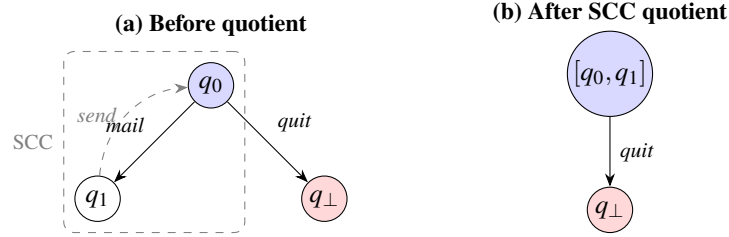


Figure 2: SMTP session type $\mu X. \&\{mail : \&\{send : X\}, quit : \mathbf{end}\}$. (a) Before quotient: the recursion redirects *send* from the placeholder to q_0 , creating a cycle $q_0 \rightarrow q_1 \rightarrow q_0$ (dashed). (b) After SCC quotient: the cycle collapses into $[q_0, q_1]$, yielding a 2-element chain.

Top absorption on SMTP. Consider the SMTP session type (Figure 2):

$$S_{smtp} = \mu X. \&\{mail : \&\{send : X\}, quit : \mathbf{end}\}$$

The body is a branch with $n = 2$. Before recursion, the state space has three states: q_0 (branch entry), q_1 (after *mail*, before *send*), and $q_\perp$ (terminal). The placeholder for X is the target of the *send* transition from q_1 .

The recursion redirects the *send* transition to the body's initial state: $q_1 \xrightarrow{send} q_0$. The placeholder is removed. Now q_0 and q_1 are mutually reachable ($q_0 \xrightarrow{mail} q_1 \xrightarrow{send} q_0$), forming an SCC. By Lemma 11, the upward-closed set $D = \{q_0, q_1\}$ collapses into a single class $[q_0, q_1]$. Since $q_\perp \notin D$ (it is minimal), the quotient retains both extrema: the result is a 2-element chain $[q_0, q_1] > q_\perp$ — a bounded lattice. Verified by the Python tool: 3 states, 3 transitions, $SCC = \{q_0, q_1\}$.

3.4 Parallel Composition

Parallel composition is the only constructor introducing a product structure rather than a sum.

Lemma 14 (Parallel). *If $\mathcal{L}(S_i)/\equiv$ is a bounded lattice for each $i \in \{1, \dots, n\}$, then $\mathcal{L}(S_1 \parallel \dots \parallel S_n)/\equiv$ is a bounded lattice.*

Proof. By the parallel case of Definition 3 and the independence of the n branches (Termination ensures every branch reaches **end**; Parallel closedness ensures no shared variables), the product inherits SCC structure componentwise: $\mathcal{L}(S_1 \parallel \dots \parallel S_n)/\equiv \cong \prod_i \mathcal{L}(S_i)/\equiv_i$. The product of finitely many bounded lattices is itself a bounded lattice (Section 2.2), illustrated for $n = 2$ in Figure 3. $\square$

3.5 The Reticulate Theorem and Realisability

The per-constructor lemmas now compose into the headline result, followed by tightness and necessary realisability conditions.

Theorem 15 (Reticulate Theorem). *For every well-formed session type S (including equation systems), $\mathcal{L}(S)/\equiv$ is a bounded lattice.*

Proof. By structural induction on S , applying the per-constructor lemmas Lemmas 8, 10 and 12 to 14. Each case preserves the bounded-lattice property, completing the induction. $\square$

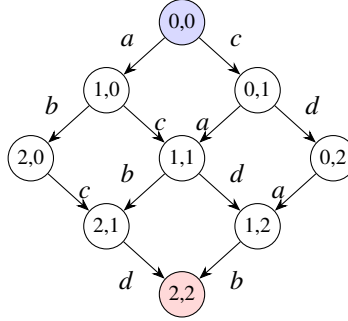


Figure 3: Product lattice $\mathcal{L}(\&\{a : \&\{b : \mathbf{end}\}\} \parallel \&\{c : \&\{d : \mathbf{end}\}\})$. The 9 states form a 3×3 grid lattice with componentwise ordering.

Corollary 16 (Termination tightness). *The **Termination** clause is tight: relaxing the exit-path condition yields types whose SCC-quotiented state spaces are not lattices.*

A witness: $\&\{a : \mathbf{end}, b : \mu X. \&\{a : X\}\}$ violates **Termination** — from the body root $\&\{a : X\}$, the only outgoing transition unfolds X back to the same root, so no path to $\mathbf{end}$ exists. After SCC quotient the recursive branch collapses to a single class with no outgoing transition to a terminal, leaving no bottom element reachable from all states.

Proposition 17 (Necessary conditions for realisability). *If a finite bounded lattice L is the reticulate $\mathcal{L}(S)/\equiv$ of some well-formed session type S , then:*

1. **Top out-degree.** $\top$ has out-degree ≥ 1 in the Hasse diagram (from the outermost constructor).
2. **Element dichotomy.** Every non-extremal element is either a join node (reached from $\top$ by a unique path of choice constructors) or a product element (a tuple from a $\parallel$ sub-lattice).
3. **Product decomposition.** Every product sub-lattice is a direct product of reticulates.

Proof. Conditions (i)–(iii) are mechanised; see Table 1. □

4 Duality and Subtyping as Lattice Consequences

The lattice structure has two consequences for class session types: duality is invisible at the lattice level (Section 4.1); width subtyping reduces to lattice embedding (Section 4.2). On *channel* session types, duality is operational, swapping $\&/\oplus$ together with $?/!$ [14]; for *class* session types, the residual polarity swap preserves lattice structure (Proposition 18).

4.1 Duality Preserves Lattice Isomorphism

The *dual* of a session type swaps branch and selection: $\mathit{dual}(\&\{m_i : S_i\}) = \oplus\{m_i : \mathit{dual}(S_i)\}$ and vice versa; all other constructors ($\mathbf{end}$, $\parallel$, $\mu X. \cdot$) are covariant. Duality is an involution: $\mathit{dual}(\mathit{dual}(S)) = S$.

Proposition 18 (Duality isomorphism). *For every well-formed session type S , there is a bounded-lattice isomorphism*

$$\mathcal{L}(S)/\equiv \cong \mathcal{L}(\mathit{dual}(S))/\equiv.$$

Proof. By structural induction on S . The base case (**end**) is trivial since duality is the identity. For choice, branch and selection produce identical state spaces (Lemma 9), so polarity reversal is invisible at the lattice level. For parallel, recursion, and named definitions, duality distributes through the constructors, and the inductive isomorphisms on sub-types compose into one on the whole. $\square$

Proposition 18 says that polarity is invisible at the lattice level: S and $dual(S)$ produce the same lattice structure. For class session types, only the object’s protocol S is explicit; the dual $dual(S)$ specifies the implicit second-party (client) protocol, and these two faces share the same lattice structure. Both tools verify $\mathcal{L}(S)/\equiv \cong \mathcal{L}(dual(S))/\equiv$ for all 108 benchmarks. The Lean 4 mechanisation proves $dual(dual(S)) = S$ on the AST; the state-space isomorphism follows from the constructor-wise argument above (see Table 1).

4.2 Subtyping as Lattice Embedding

The lattice structure has a concrete consequence for subtype checking. Gay and Hole [13] define a coinductive subtyping relation $\leq_{GH}$ on session types as the greatest fixed point of a monotone operator on type-indexed binary relations. In words, $S_1 \leq_{GH} S_2$ means the subtype S_1 can be substituted wherever a S_2 -typed object is expected — a branch subtype offers *more* methods (covariant width); a selection subtype emits *fewer* labels (contravariant width); parallel is congruent; recursion is closed under one-step unfolding. We refer to [13] for the formal inference rules.

The contravariance of selection width is the *dual* of branch covariance: by Proposition 18, swapping polarity preserves the lattice, so the selection case follows from the branch case applied to the dual types. Since branch and selection produce *identical state spaces* (Section 3), the embedding relationship depends on which type has more top-level choices — not on the polarity.

Definition 19 (Lattice embedding). An *embedding* of a bounded lattice L_1 into L_2 is an injective map $\phi : L_1 \rightarrow L_2$ that preserves and reflects order: $a \leq b$ iff $\phi(a) \leq \phi(b)$. Every embedding preserves meets and joins.

Proposition 20 (Width–embedding correspondence). *For all well-formed non-recursive session types related by width subtyping (shared-label continuations coincide, $S_i = S'_i$ for $i \leq k$):*

1. **Branch width.** *If $\&\{m_1:S_1, \dots, m_n:S_n\} \leq_{GH} \&\{m_1:S'_1, \dots, m_k:S'_k\}$ with $k \leq n$, then $\mathcal{L}(S_2)/\equiv$ embeds into $\mathcal{L}(S_1)/\equiv$ (supertype into subtype).*
2. **Selection width.** *If $\oplus\{l_1:S_1, \dots, l_n:S_n\} \leq_{GH} \oplus\{l_1:S'_1, \dots, l_k:S'_k\}$ with $n \leq k$, then $\mathcal{L}(S_1)/\equiv$ embeds into $\mathcal{L}(S_2)/\equiv$ (subtype into supertype).*

Proof. Without recursion the state space is acyclic, so the SCC quotient is the identity and no states merge. For width subtyping the shared continuations coincide, so the supertype’s reticulate is exactly the subtype’s restricted to the shared labels; the extra labels contribute fresh sub-lattices disjoint above $q_\perp$ (Lemma 10). This inclusion is injective, order-preserving and order-reflecting. Selection-width follows by duality (Proposition 18). $\square$

The two clauses share one principle: the side with fewer top-level choices has the smaller lattice, and it embeds into the larger. The flip in direction — supertype-into-subtype for $\&$, subtype-into-supertype for $\oplus$ — is not a property of the lattice (which by Lemma 9 ignores polarity) but of the subtyping relation $\leq_{GH}$: covariant in $\&$ -width, contravariant in $\oplus$ -width. The lattice picture thus unifies covariance and contravariance into a single size statement.

Proposition 20 characterises *width* subtyping. The full Gay–Hole relation $\leq_{GH}$ also includes *depth* subtyping: $S_i \leq_{GH} S'_i$ for shared labels. In the lattice picture, each shared continuation $S_i \leq_{GH} S'_i$ contributes its own embedding between $\mathcal{L}(S_i)/\equiv$ and $\mathcal{L}(S'_i)/\equiv$, oriented by the polarity of S_i (clause 1 or 2); as polarity alternates with depth these run in both directions — some subtype-into-supertype, some the reverse — a *family* of embeddings, not a single map. Thus the full relation decomposes as: width determines *which* sub-lattices participate in the embedding, and depth determines *how* corresponding sub-lattices relate. A complete lattice-theoretic characterisation of $\leq_{GH}$ (including the interaction of width and depth with recursion) is the subject of ongoing work.

Both tools implement the embedding check and verify the correspondence on all 108 benchmarks (47 with recursion); Section 5 demonstrates the construction concretely.

5 Case Study: A Concurrent File Protocol

We demonstrate Theorem 15, Proposition 18, and Proposition 20 on a sequential file reader, then recover its writer counterpart and compose the two in parallel into the canonical concurrent read/write File.

Construction. The FileReader protocol — a single thread reading from open to close, with each read returning *data* until *eof* — is

$$S_{\text{FileReader}} = \&\{\text{open} : \mu X. \&\{\text{read} : \oplus\{\text{data} : X, \text{eof} : \text{Close}\}\}, \quad \text{Close} = \&\{\text{close} : \mathbf{end}\}.$$

The bottom-up construction matches subterms to constructor lemmas:

Subterm	Lemma
end	Lemma 8
$\text{Close} = \&\{\text{close} : \mathbf{end}\}$	Lemma 10, $n=1$; Lemma 13
$\oplus\{\text{data} : X, \text{eof} : \text{Close}\}$	Lemma 10, $n=2$
$\&\{\text{read} : \dots\}$	Lemma 10, $n=1$
$\mu X. \dots$	Lemma 12, Lemma 11
$\&\{\text{open} : \dots\}$	Lemma 10, $n=1$

The recursion step makes q_1, q_2 mutually reachable, so Lemma 11 collapses them into the class $[q_1, q_2]$: the SCC captures the *read/data loop*, the protocol cycling between read and selection indefinitely before choosing *eof*. By Theorem 15, the composition witnesses that $\mathcal{L}(S_{\text{FileReader}})/\equiv$ is the bounded lattice $q_0 > [q_1, q_2] > q_3 > q_\perp$ shown in Figure 4(b).

Duality. The dual swaps every choice polarity:

$$\text{dual}(S_{\text{FileReader}}) = \oplus\{\text{open} : \mu X. \oplus\{\text{read} : \&\{\text{data} : X, \text{eof} : \oplus\{\text{close} : \mathbf{end}\}\}\}\}.$$

Each choice flips agency: methods that $S_{\text{FileReader}}$ *offers* are *invoked* by $\text{dual}(S_{\text{FileReader}})$, and outcomes the file *internally selects* are *received* by the dual — the same protocol read from the opposite endpoint. By Lemma 9, replaying Definition 3 yields the same Hasse diagram (Figure 4(b)), witnessing Proposition 18.

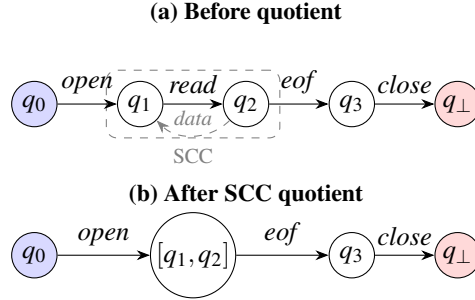


Figure 4: $\mathcal{L}(S_{\text{FileReader}})$: (a) before quotient, with the recursion edge $q_2 \rightarrow q_1$ dashed; (b) after the SCC quotient $[q_1, q_2]$, a 4-element chain.

Subtyping. Consider a richer file protocol that also lets a client read metadata and close without entering the read loop:

$$S_{\text{FileReader}}^+ = \&\{open : \mu X. \&\{read : \oplus\{data : X, eof : \&\{close : \mathbf{end}\}\}\}, stat : \&\{close : \mathbf{end}\}\}$$

By branch covariance (Gay–Hole width subtyping), $S_{\text{FileReader}}^+ \leq_{GH} S_{\text{FileReader}}$: the subtype offers *more* top-level methods. Applying Definition 3, the outer branch widens the top q_0 with a second transition $q_0 \xrightarrow{stat} q_4$, where $q_4 = \&\{close : \mathbf{end}\}$ is a fresh state reached *outside* the read loop; the shared *open* arm reproduces Steps 1–6 unchanged. After SCC quotient the subtype’s reticulate is the five-element lattice

$$q_0 > [q_1, q_2] > q_3 > q_{\perp}, \quad q_0 > q_4 > q_{\perp},$$

with top q_0 and the two arms meeting at $q_{\perp}$ (q_3, q_4 incomparable). The supertype’s 4-chain embeds into it as the *open* arm, q_4 lying outside the image — a strict, order-preserving and -reflecting embedding. This type is recursive, so it lies outside Proposition 20; here the recursion sits in the shared *open* continuation, so the embedding survives.

FileWriter. The symmetric writer arises by renaming *read/data* to *write/ack*:

$$S_{\text{FileWriter}} = \&\{open : \mu X. \&\{write : \oplus\{ack : X, eof : \&\{close : \mathbf{end}\}\}\}\}.$$

The construction is invariant under method renaming, so $\mathcal{L}(S_{\text{FileWriter}}) \cong \mathcal{L}(S_{\text{FileReader}})$.

File. The canonical concurrent read/write File bundles both inside one object: $S_{\text{File}} \triangleq S_{\text{FileReader}} \parallel S_{\text{FileWriter}}$. Two threads sharing only a filename each instantiate one arm independently, so two opens and two closes on the same logical file are not errors but transitions on disjoint arms. The reticulate is the componentwise product $\mathcal{L}(S_{\text{FileReader}}) \times \mathcal{L}(S_{\text{FileWriter}})$ (Lemma 14).

6 Implementation, Mechanisation, and Empirical Validation

Implementation. We have implemented the state-space construction and lattice checking in two independently developed tools targeting distinct deployment surfaces: a Python 3.12 CLI for standalone protocol analysis, and a Java 21 annotation processor that performs the check at compile time on `@Session`-annotated code. Both are thoroughly tested and agree on all 108 benchmarks. The pipeline is identical

Table 1: Lean 4 mechanisation: import closure of `reticulate_lattice` (11 modules) plus four auxiliary propositions; 15 modules total, zero `sorry`, no coinduction.

Role	Lean 4 module
Directed graphs	<code>Reticulate/DirectedGraph.lean</code>
Reachability	<code>Reticulate/Reachability.lean</code>
SCC quotient	<code>Reticulate/SCC.lean</code>
AST (Section 2.1)	<code>Reticulate/Spec/SessionType.lean</code>
Free variables	<code>Reticulate/Spec/FreeVars.lean</code>
Termination check	<code>Reticulate/Spec/Termination.lean</code>
4-clause well-formedness	<code>Reticulate/Spec/WellFormed.lean</code>
Builder (Definition 3)	<code>Reticulate/Spec/StateSpace.lean</code>
Edge construction	<code>Reticulate/Spec/StateSpaceEdges.lean</code>
Reachability lemmas	<code>Reticulate/Spec/Reachability.lean</code>
Theorem 15	<code>Reticulate/Spec/StateSpaceLattice.lean</code>
Proposition 18	<code>Reticulate/Spec/Duality.lean</code>
Proposition 20	<code>Reticulate/Spec/SubtypingEmbedding.lean</code>
Proposition 17	<code>Reticulate/Spec/Realizability.lean</code>
Divergent-recursion counterexample	<code>Reticulate/Counterexample.lean</code>

in both tools: a parser converts the session-type source (Java `@Session` annotations or text input) to an AST; a builder applies Definition 3 clause-by-clause to produce the labelled state space; the SCC quotient is computed via Tarjan’s algorithm [29]; and an all-pairs check verifies $\wedge$ and $\vee$ existence on the quotient. The lattice check is $O(n^2 \cdot m)$ where $n = |Q|$ and $m = |\delta|$, completing in under 100 ms on commodity hardware for all benchmarks (median 4 ms, maximum 87 ms for the largest benchmark with $|Q| = 41$).

Test generation. A direct application of the lattice goes beyond verification: automated protocol-conformance test generation. Both implementations expose a test-generation mode that emits JUnit 5 cases in three categories: valid paths (chains from $q_\top$ to $q_\perp$, enumerated by bounded DFS), violations (disabled methods at each reachable state), and incomplete prefixes (proper prefixes of valid paths). The lattice’s boundedness guarantees a finite test suite; bottom-reachability (Proposition 7) guarantees every valid path terminates. Non-terminating types are accepted under the same three-mode flag `--non-termination={error,warn,off}` as the lattice check; modes, default, and exact wording are pinned identical across both tools by mirrored parity test suites. Valid paths are then empty, but violations and incomplete prefixes remain. Both tools agree on the generated test sets across the 108 benchmarks.

Mechanisation. The paper’s results have been mechanically verified in Lean 4 [22] using Mathlib [30]. Theorem 15 is mechanised as a universal theorem `reticulate_lattice` quantifying over every well-formed session type, with no extra hypothesis; Proposition 18, Proposition 20, and Proposition 17 are backed by dedicated modules in the `Reticulate.Spec` namespace. The full paper-faithful artefact comprises 15 modules with zero `sorry` and zero `admit` (Table 1).

Empirical validation. The 108-protocol benchmark suite plays two distinct roles: (i) it tests that the Python and Java implementations *conform* to the theorem on real protocols (tooling conformance); and (ii) it characterises the theorem’s empirical scope — every protocol in the suite is well-formed and produces a lattice, and the 86%/14% distributive split (below) gives a first empirical picture of `reticulate`

structure across application domains. The 108 protocols (Table 2) draw from *literature benchmarks* (Java Iterator, SMTP, HTTP, JDBC, OAuth 2.0, Two-Buyer, and others from the Mungo/StMungo/Scribble literature), *industrial protocols* (MCP, A2A, TLS, Saga Orchestrator, WebSocket, gRPC), and *synthetic benchmarks* systematically covering edge cases (deep recursion, wide branching $n \geq 10$, nested parallel, mixed polarity).

Table 2: Benchmark summary by source.

Source	Protocols	Distributive	Non-dist.
Literature	34	32	2
Industrial	26	18	8
Synthetic	48	43	5
Total	108	93	15

All 108 protocols produce bounded lattices under SCC quotient. We classify each lattice in the Birkhoff hierarchy [4]: 93 (86%) are *distributive* (no forbidden N_5 or M_3 sublattice) and 15 (14%) are non-distributive — 13 containing N_5 (pentagon) and 3 containing M_3 (diamond); one protocol (Hex) contains both. All product lattices from parallel composition are distributive, consistent with the lattice-theoretic result that the product of distributive lattices is distributive. This classification is mechanised in Lean 4 with a proof (by `decide`) that the distributive law fails on N_5 .

Two structural patterns account for all 15 non-distributive cases: N_5 from *depth-unequal choice* (branches of a choice constructor terminating at different depths yield a 3-chain plus side-branch) and M_3 from *wide choice* ($n \geq 3$ branches reconverging at $q_\perp$ yield three pairwise-incomparable atoms; binary choice produces at most M_2 , which is distributive). Both patterns are polarity-symmetric (Proposition 18); the empirical clustering of N_5 in polarity-mixed nestings and M_3 in wide branchings (Table 3) reflects regularities of real protocols, not structural requirements. Characterising precisely which session types yield distributive reticulates remains open.

We additionally enumerate all well-formed terminating session types up to depth 3 with label sets $\{a, b\}$, confirming that every such type produces a lattice — further evidence for universality.

7 Related Work

Session types. Session types were introduced by Honda et al. [15] for binary communication and extended to multiparty settings [16]; Vasconcelos [32], Hüttel et al. [17], and Ancona et al. [2] provide foundational treatments and comprehensive surveys. Honda et al.’s formulation is a process calculus with explicit $?T/!T$ directionality and delegation; ours is an object-protocol specification on a single shared object, with $\parallel$ over n participants on that object rather than between independent processes. Session types for objects were developed by Vasconcelos et al. [33], Dezani-Ciancaglini et al. [12], and Gay et al. [14], with tool support from MUNGO and STMUNGO [19, 20]. MUNGO uses named definitions as the primary design mechanism; our grammar adopts the same notation. By contrast, our $\parallel$ constructor has no counterpart in typestate tools. Gay and Hole [13] established subtyping for session types (revisited as lattice embedding in Section 4.2). Scalas and Yoshida [27] showed that binary session types can be simplified by reducing to a minimal core. Across the work above, the focus is on the type algebra; we study the algebraic structure of the *state spaces* induced by individual session types.

Table 3: The 15 non-distributive benchmarks. 13 contain N_5 (pentagon); 3 contain M_3 (diamond).

Protocol	$ Q $	Type	Cause
TLS Handshake	13	N_5	branch-inside-selection (HELLO_RETRY)
Two-Phase Commit	7	N_5	branch-selection nesting
Ki3 Onboarding	32	N_5	nested branch-selection
Ki3 CI/CD Pipeline	41	N_5	4-stage branch-selection nesting
Quantum Measurement	9	N_5	crossed branch-selection
SSH Handshake	8	N_5	branch-selection nesting
Alternative Splicing	7	M_3	three branches sharing outcomes
QCD Gluon Exchange	6	M_3	three-way symmetric reconvergence
Hex (2×2)	10	both	game-tree reconvergence + nesting
Plus 6 synthetic benchmarks with analogous patterns.			

Intersection/union session types. Padovani [24, 23] and Acay and Pfenning [1] develop a *type-algebra* lattice where internal and external choice are replaced by intersection $\wedge$ and union $\vee$, yielding a semantic subtyping lattice $\langle 2^{\mathcal{P}}, \subseteq \rangle$ of closed process-set denotations. Our lattice is orthogonal: it lives *inside* a single type, on its reachable state space. Padovani [24] (p. 80) conjectures distributivity of his process-set lattice and leaves it open; our 86%/14% distributive split on 108 benchmarks (Section 6) is a first empirical probe of the analogous question on reachability lattices, with a formal bridge left to future work. Barbanera and de’Liguoro [3] similarly develop a sub-behaviour preorder on client/server session types via LTS compliance, but prove duals-as-minima rather than a lattice. Strom and Yemini’s 1986 tpestate proposal [28] required tpestates of each type to form a meet-semilattice — the closest historical ancestor. But their lattice captures abstract-interpretation dataflow, not protocol reachability; in particular, our $\parallel$ constructor is what forces completion to a *full* lattice.

Lattices in programming languages. Lattice theory underlies abstract interpretation [7] and is the standard algebraic vocabulary for order in PL [4, 11]. We show that *behavioural* types produce lattices at the state-space level, a distinct setting.

Linear logic and session types. Caires and Pfenning [5] established a Curry–Howard correspondence between session types and intuitionistic linear logic; Wadler’s [34] classical version (“propositions as sessions”) links negation to session-type duality, complementing our state-space duality (Proposition 18). Toninho, Caires, and Pfenning [31] extended the correspondence to higher-order sessions; the lattice-theoretic perspective on higher-order types remains open.

Subtyping. Gay and Hole [13] defined subtyping for session types; Dardha et al. [10] revisited it in a pi-calculus setting. Padovani [25] extended subtyping to multiparty types with fairness; his coinductive characterisation operates on infinite trees, complementary to our finite quotient embedding. Castagna and Frisch [6] developed semantic subtyping using set-theoretic models. Our characterisation of subtyping as lattice embedding (Section 4.2) provides an algebraic alternative to the coinductive, fair, and set-theoretic approaches.

Multiparty session types. Honda, Yoshida, and Carbone [16] introduced multiparty session types (MPST), where a global type is projected onto local types for each role. Our Reticulate Theorem applies to local types obtained by projection; whether projection itself is a lattice morphism between global and

local reticulates is future work. Dagnino, Giannini, and Dezani-Ciancaglini [8] recently developed de-confined global types in a coinductive, process-calculus setting proving progress and deadlock-freedom; our setting is complementary — single object, finite-state (μ -unfolded), specification-based — proving lattice structure rather than progress.

Process algebra. Our $\parallel$ operates on a *single shared object* rather than independent processes [21, 26], requiring product lattice construction. Kobayashi [18] and Dardha and Gay [9] use type systems for deadlock-freedom; our lattices complement this by making all interleavings explicit, guaranteeing that every concurrent path reaches $\perp$.

8 Conclusion

We have proved that the state space of every well-formed session type, quotiented by strongly connected components, forms a bounded lattice (Theorem 15). The proof is constructive and modular, with one lemma per constructor; the grammar is parsimonious (six type constructs, three n -ary), and well-formedness reduces to four clauses (Definition 1), three standard in the session-type literature and one (termination) specific to our setting. We validated the theorem on 108 benchmark protocols using two independently developed, thoroughly tested tools, and mechanically verified the core proof in Lean 4 (15 modules, zero sorry; Table 1).

This lattice structure — the *reticulate* — opens several directions. The *distributivity classification* (86% distributive, 14% non-distributive) suggests finer algebraic structure: distributive lattices admit unique prime decompositions (Birkhoff’s theorem), which could yield canonical forms for session types. The converse — characterising which bounded lattices arise as reticulates — is addressed by Proposition 17; a complete characterisation remains open. A Galois connection between an implementation’s concrete state space and a session type’s reticulate would yield a sound abstract interpretation for protocol conformance [7]. Our duality is structural — it identifies session types whose state spaces coincide via polarity erasure (Lemma 9). Richer semantic dualities, between method pairs (e.g., reader/writer, producer/consumer) or with directional payload tags, fit naturally since the construction is invariant under method renaming; [14]’s channel duality, viewed through this lens, configures the two endpoints as a 2-arm parallel composition with mutually dual arms (modulo $?/!$), and a formal translation including the $?/!$ half is open. Three grammar extensions preserve lattice structure and are the subject of companion work: first-class sequential composition, synchronous parallel, and replication.

The Reticulate Theorem suggests that session types are more structured than previously recognised: the lattice emerges from the syntax rather than being imposed by the theory.

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A Detailed Proofs

This appendix collects the manual proofs of the lemmas and theorems of Section 3, together with the structural arguments for Proposition 18 and Proposition 20. Each result is also mechanised; per-result Lean module pointers are given in the main-body proof stubs.

Proof of Proposition 7

Proof. By induction on the construction: each case introduces $q_\top$ and $q_\perp$ (End: single state; choice: fresh $q_\top$ plus shared $q_\perp$; recursion: back-edges preserve extrema in the quotient; parallel: componentwise extrema). $\square$ $\square$

Proof of Lemma 9

Proof. Immediate from the construction: both create a fresh $q_\top$ with n transitions to sub-state-space entries, sharing $q_\perp$. Polarity is a typing discipline, not a structural property of the state space. $\square$ $\square$

Proof of Lemma 8

Proof. The state space $\mathcal{L}(\mathbf{end})$ consists of a single state q that is both initial and terminal: $q_\top = q_\perp = q$. The SCC quotient $\mathcal{L}(\mathbf{end})/\equiv$ is the one-element poset $\{[q]\}$, which satisfies all lattice axioms: $[q]$ is both top and bottom, and $[q] \wedge [q] = [q] \vee [q] = [q]$. $\square$ $\square$

Proof of Lemma 10

Proof. The construction creates a fresh initial state q_0 with transitions to each sub-state-space entry, and a shared terminal $q_\perp$. For states in the same sub-lattice, meets and joins exist by induction. For states $a \in \mathcal{L}(S_i)$ and $b \in \mathcal{L}(S_j)$ ($i \neq j$): the sub-state-spaces are disjoint above $q_\perp$, so any upper bound of $\{a, b\}$ must reach both sub-spaces and therefore must be q_0 (the only such state); hence $a \vee b = q_0$. Dually, any lower bound of $\{a, b\}$ must be reachable from both, hence must lie at or below $q_\perp$; since $q_\perp$ is least, $a \wedge b = q_\perp$. See Figure 1 for an illustration. $\square$ $\square$

Proof of Lemma 11

Proof. Since D is a proper upward-closed subset of L , $\perp \notin D$ (otherwise upward-closure would force $D = L$), so L' retains both extrema. For $a, b \in L \setminus D$: the meet $a \wedge_L b$ lies in $L \setminus D$ (the complement of an upward-closed set is downward-closed), so it is preserved. For the join: if $a \vee_L b \in L \setminus D$, it is preserved. If $a \vee_L b \in D$, then $a \vee_{L'} b = \top$: any non- $\top$ upper bound $u \in L \setminus D$ would satisfy $u \geq a \vee_L b \in D$, placing $u \in D$ since D is upward-closed — contradicting $u \in L \setminus D$. $\square$ $\square$

Proof of Lemma 12

Proof. Adding back-edges from each placeholder p_X to the initial state $q_\top^S$ makes these states mutually reachable with $q_\top^S$, hence they are collapsed into the same equivalence class $[q_\top^S]$. The set D of quotient classes absorbed into $[q_\top^S]$ is upward-closed: if $[s] \in D$ and $[s] \leq [r]$, then $r \twoheadrightarrow s$ (since $[r] \geq [s]$), $s \twoheadrightarrow q_\top^S$ (since $[s] \in D$), and $q_\top^S \twoheadrightarrow r$ (as initial state); so r and $q_\top^S$ are mutually reachable, giving $[r] \in D$. By the Termination clause, $[q_\perp]$ is reachable from $[q_\top^S]$ in $\mathcal{L}(\mu X.S)/\equiv$, so $[q_\perp] \notin D$; hence D is a *proper* upward-closed subset of the body quotient. By Lemma 11, collapsing D into a single top element yields a bounded lattice. $\square$ $\square$

Proof of Lemma 13

Proof. The construction connects the k named sub-state-spaces by redirecting placeholder transitions for each name X_j to $q_{\top}^{S_j}$. Non-recursive references (acyclic names) are effectively inlined and preserve lattice structure trivially. Cyclic references create back-edges that merge states into SCCs, exactly as in the single-recursion case. The set of quotient classes absorbed into each SCC is upward-closed by the same argument as Lemma 12. Applying Lemma 11 iteratively (once per SCC created by cyclic references) yields a bounded lattice. The SCC quotient is determined by the reachability graph alone (Definition 6); independence from the absorption order is a confluence property mechanised as `NamedDefinitions.absorption_order_independent`. $\square$ $\square$

Proof of Lemma 14

Proof. By the parallel case of Definition 3 and the independence of the n branches (Termination ensures every branch reaches **end**; Parallel closedness ensures no shared variables), the product inherits SCC structure componentwise — an SCC in the product is a product of SCCs from each component:

$$\mathcal{L}(S_1 \parallel \dots \parallel S_n) / \equiv \cong \prod_{i=1}^n (\mathcal{L}(S_i) / \equiv_i).$$

The product of finitely many bounded lattices is a bounded lattice (meets and joins are componentwise). See Figure 3 for an illustration with $n = 2$. $\square$ $\square$

Proof of Theorem 15

Proof. By structural induction on S , applying Lemmas 8, 10 and 12 to 14 case-by-case. Each case preserves the bounded lattice property, completing the induction. $\square$ $\square$

Proof of Proposition 18

Proof. By structural induction on S .

Base case ($S = \mathbf{end}$): $\mathit{dual}(\mathbf{end}) = \mathbf{end}$, so the state spaces are identical.

Choice ($S = \&\{m_1:S_1, \dots, m_n:S_n\}$): $\mathit{dual}(S) = \oplus\{m_1:\mathit{dual}(S_1), \dots, m_n:\mathit{dual}(S_n)\}$. By the construction in Section 3, branch and selection produce identical state spaces: both create a fresh $q_{\top}$ with n transitions to sub-state-space entries, sharing $q_{\perp}$. The construction depends only on the set of continuations, not on the polarity. By induction, $\mathcal{L}(S_i) / \equiv \cong \mathcal{L}(\mathit{dual}(S_i)) / \equiv$ for each i . The n sub-isomorphisms, together with the identity on $q_{\top}$ and $q_{\perp}$, yield an isomorphism of the whole choice state space.

Recursion ($S = \mu X.S'$): $\mathit{dual}(\mu X.S') = \mu X.\mathit{dual}(S')$. The recursion construction redirects placeholder transitions to the body's initial state. Since dual acts on the body's constructors (swapping $\&$ and $\oplus$) but preserves the placeholder and entry-state structure, the same back-edges are created in both $\mathcal{L}(\mu X.S')$ and $\mathcal{L}(\mu X.\mathit{dual}(S'))$, yielding isomorphic SCC quotients by induction on the body.

Parallel ($S = S_1 \parallel \dots \parallel S_n$): dual distributes over parallel, so $\mathit{dual}(S) = \mathit{dual}(S_1) \parallel \dots \parallel \mathit{dual}(S_n)$. By induction each factor is isomorphic; the product of isomorphic lattices is isomorphic.

Named definitions ($D = S, X_1 = S_1, \dots, X_k = S_k$): $\mathit{dual}(D) = \mathit{dual}(S), X_1 = \mathit{dual}(S_1), \dots, X_k = \mathit{dual}(S_k)$. Since dual acts on the bodies but preserves the name-reference structure, the same redirections are performed in both $\mathcal{L}(D)$ and $\mathcal{L}(\mathit{dual}(D))$, yielding isomorphic SCC quotients by induction on each S_i . $\square$ $\square$

Proof of Proposition 20

Proof. Gay and Hole [13] define subtyping on recursive types coinductively as the greatest fixpoint of a monotone operator: assume the pair (S_1, S_2) is in the relation, unfold recursive types one step, and check the structural conditions.

Reduction from coinduction to finite induction. The coinductive derivation unfolds recursive types repeatedly, yielding an infinite derivation tree. However, each unfolding $\mu X.S \rightarrow S[X := \mu X.S]$ corresponds to a back-edge within the same SCC class (Lemma 12). Since the SCC quotients $\mathcal{L}(S_1)/\equiv$ and $\mathcal{L}(S_2)/\equiv$ are finite DAGs, the derivation visits only finitely many distinct quotient-class pairs. Any derivation step that revisits a pair $([s_1], [s_2])$ already encountered is an implicit appeal to the coinductive hypothesis — on the finite quotient, this corresponds to closing a cycle. Therefore the infinite coinductive derivation factors through a finite set of quotient-class pairs, and we may reason by induction on the DAG structure of the quotient.

Embedding construction. For branch width: $\&\{m_1:S_1, \dots, m_n:S_n\} \leq_{GH} \&\{m_1:S'_1, \dots, m_k:S'_k\}$ with $k \leq n$ requires $S_i \leq_{GH} S'_i$ for each $i \leq k$ (covariant depth). By induction on the quotient DAG, each $\mathcal{L}(S'_i)/\equiv$ embeds into $\mathcal{L}(S_i)/\equiv$ via some φ_i . Construct the embedding φ of the supertype lattice into the subtype lattice by: $\varphi([q_\top]) = [q_\top]$, $\varphi([q_\perp]) = [q_\perp]$, and for each quotient class $[s]$ in $\mathcal{L}(S'_i)/\equiv$, $\varphi([s]) = \varphi_i([s])$. This is well-defined because the k sub-lattices of the supertype are disjoint above $[q_\perp]$ in the quotient, so the φ_i do not conflict; it is order-preserving and order-reflecting by the inductive hypothesis on each component and the shared extrema. Selection width is analogous by duality (Proposition 18).

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